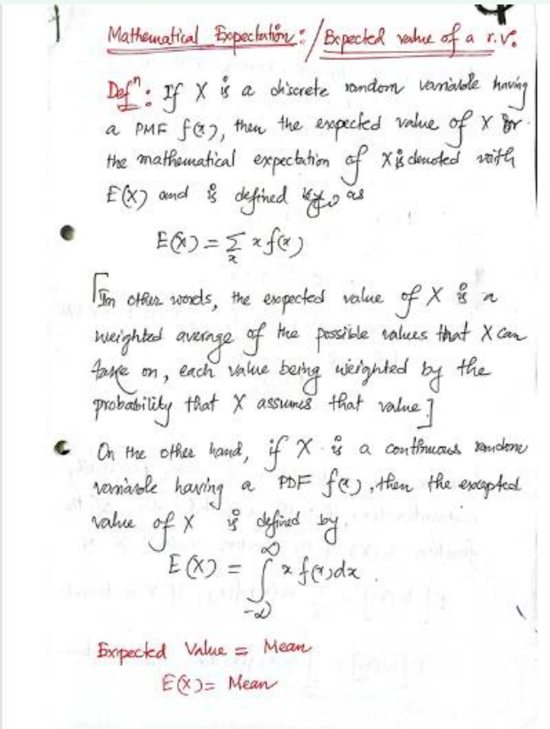
I will give you screenshots of some handwritten math notes. You will explain me that topic in the way it is in the note and give intuition for better understanding. If it is a math or derivation, you will explain it step by step clearly. OK?

Absolutely—send the screenshots and I've got you.

Here's how I'll handle each page:

1. **Rewrite the content cleanly (same symbols/notation as your notes).**
2. **Step-by-step derivation/solution** with every algebra move shown.
3. **Intuition/visual meaning** (why the steps make sense, what the formula "wants to say").
4. **Common pitfalls & quick checks** so you can self-verify.
5. **Mini summary/cheat-sheet** at the end.

If you have multiple images, please send them **in order** (1, 2, 3…). If any part is especially confusing, tell me which line you want extra focus on. I can also add a brief **Bangla explanation** alongside the English if you'd like.



Mathematical Expectation: / Expected value of a r.v.:

Def^n: If X is a discrete random variable having a PMF $f(x)$, then the expected value of X or the mathematical expectation of X is denoted with $E(X)$ and is defined by as

$$E(X) = \sum_x x f(x)$$

[In other words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes that value]

On the other hand, if X is a continuous random variable having a PDF $f(x)$, then the expected value of X is defined by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Expected Value = Mean

$E(X)$ = Mean

Perfect—thanks for the screenshot! Let's carefully go through it.

✦ Topic: Mathematical Expectation / Expected Value of a Random Variable

1. Definition for Discrete Random Variables

If X is a **discrete random variable** with probability mass function (PMF) $f(x)$, then the **expected value** (or mean) of X is:

$$E(X) = \sum_x x f(x)$$

✅ **Meaning**:

- Take every possible value of X (denoted as x).
- Multiply it by the probability of that value occurring $f(x)$.
- Add (sum) them all up.

This is just a **weighted average**, where probabilities act as the weights.

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2. Definition for Continuous Random Variables

If X is a **continuous random variable** with probability density function (PDF) $f(x)$, then:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

✅ **Meaning**:
Instead of summing over discrete values, we integrate over the entire real line. The PDF $f(x)$ gives the density of probability, and multiplying by x and integrating gives the "average" value.

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3. Intuitive Understanding

- Expected value = **center of gravity** of the distribution.
- It tells us what value we can "expect on average" if we repeat the random experiment many times.
- For a fair die:

  $$E(X) = \frac{1+2+3+4+5+6}{6} = 3.5$$

  Even though 3.5 never appears on the die, it is the long-run average.

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4. Important Notes

- The expected value is the **mean** of the random variable.
- It may **not always be one of the possible values** (like 3.5 in the die example).
- It provides a single representative value of the distribution.

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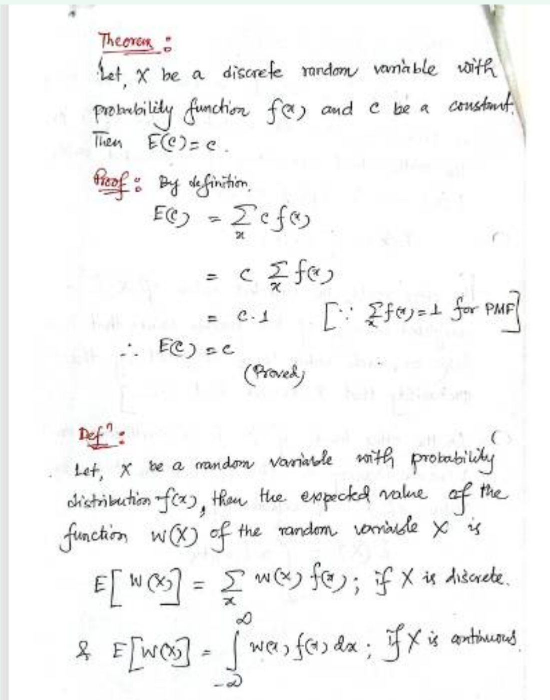
5. Summary

- **Discrete case:** $E(X) = \sum x f(x)$
- **Continuous case:** $E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$
- **Interpretation:** Weighted average / Mean.

👉 Would you like me to also show you a **step-by-step solved example** (one discrete and one continuous) to make this definition crystal clear?



Theorem :

Let, X be a discrete random variable with probability function $f(x)$ and c be a constant. Then $E(c) = c$.

Proof : By definition,

$E(c) = \sum_x c f(x)$

$= c \sum_x f(x)$

$= c \cdot 1 \quad [\because \sum f(x) = 1$ for PMF$]$

$\therefore E(c) = c$

(Proved)

Def^n :

Let, X be a random variable with probability distribution $f(x)$, then the expected value of the function $w(X)$ of the random variable X is

$E[w(X)] = \sum_x w(x) f(x)$; if X is discrete.

& $E[w(X)] = \int_{-\infty}^{\infty} w(x) f(x)\, dx$; if X is continuous

Great—this page builds on the first one. Let's go step by step.

✦ Theorem:

Statement:
If X is a discrete random variable with PMF $f(x)$, and c is a constant, then

$$E(c) = c$$

Proof:

By definition of expectation:

$$E(c) = \sum_x c \cdot f(x)$$

Factor out the constant c:

$$E(c) = c\sum_x f(x)$$

But we know that for any probability mass function (PMF):

$$\sum_x f(x) = 1$$

So:

$$E(c) = c \cdot 1 = c$$

✅ **Result:**
The expectation of a constant is just the constant itself.

Intuition:

If a random variable always takes the **same constant value** c (no randomness), then the “average” is obviously just c.
Example: If you flip a “coin” that always shows 5, then the expected value is 5.

✦ General Definition (Expected value of a function of X)

Let $W(X)$ be **any function** of the random variable X.

Then:

- For **discrete** random variable:

$$E[W(X)] = \sum_x W(x)f(x)$$

- For **continuous** random variable:

$$E[W(X)] = \int_{-\infty}^{\infty} W(x)f(x)\,dx$$

Intuition:

Instead of just finding the mean of X, sometimes we're interested in the mean of some **transformation of X**, like X^2, $\ln(X)$, or $|X|$.

Examples:

- $E(X^2) \rightarrow$ helps compute variance.
- $E(|X|) \rightarrow$ expected absolute value.
- $E(g(X)) \rightarrow$ expectation of any function of X.

This is a **powerful generalization** of expectation.

✅ **Summary of this page:**

1. $E(c) = c$ (expectation of constant = the constant).
2. For any function $W(X)$, we can compute expectation using the same formula: sum (discrete) or integral (continuous).

👉 Do you want me to **solve an example using $E[X^2]$** (variance link), to make this "function of random variable" idea clearer?

Variance: Let X be a random variable with finite mean $\mu = E(X)$, then the variance of X is denoted by $V(X)$, and is defined as

$$V(X) = E(X-\mu)^2$$

$$\text{or, } V(X) = E\left[(X - E(X))^2\right].$$

Standard Deviation: The positive square root of the variance is known as the standard deviation

i.e $\sigma = \sqrt{V(X)} = \sqrt{E(X-\mu)^2}$

So, we can write,

$$V(X) = \sigma^2$$

Theorem: Let X be a discrete random variable with probability mass function $f(x)$; then

$$V(X) = \sigma^2 = E(X-\mu)^2 = E(X^2) - \mu^2$$

Proof: By definition.

$$V(X) = \sigma^2 = E(X-\mu)^2; \text{ where } \mu = E(X)$$

$$\Rightarrow \sigma^2 = E(x^2 - 2X\mu + \mu^2)$$

$$\Rightarrow \sigma^2 = E(x^2) - 2\mu E(X) + E(\mu^2)$$

$$= E(X^2) - 2\mu . \mu + \mu^2$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$\therefore \sigma^2 = E(X^2) - \mu^2$$

where, $\boxed{E(X^2) = \sum_x x^2 f(x)}$

Excellent — now we're moving into **variance and standard deviation**. Let's carefully unpack the notes.

✦ Variance

Definition:
If X is a random variable with finite mean $\mu = E(X)$, the **variance** of X is:

$$V(X) = E\left[(X - \mu)^2\right]$$

or equivalently:

$$V(X) = E\left[(X - E(X))^2\right]$$

✦ Standard Deviation

The **standard deviation** is the **positive square root of variance**:

$$\sigma = \sqrt{V(X)} = \sqrt{E\left[(X-\mu)^2\right]}$$

Thus:

$$V(X) = \sigma^2$$

✅ So variance and standard deviation are directly linked:

- Variance = average squared deviation from the mean.
- Standard deviation = "typical" deviation from the mean (in original units).

✦ Theorem (Shortcut Formula for Variance)

For a discrete random variable X with PMF $f(x)$:

$$V(X) = \sigma^2 = E(X^2) - [E(X)]^2$$

✦ Proof (Step-by-step)

Start with definition:

$$V(X) = \sigma^2 = E\left[(X-\mu)^2\right], \quad \mu = E(X)$$

Expand the square:

$$(X-\mu)^2 = X^2 - 2X\mu + \mu^2$$

So:

$$\sigma^2 = E(X^2 - 2X\mu + \mu^2)$$

Use **linearity of expectation**:

$$\sigma^2 = E(X^2) - 2\mu E(X) + E(\mu^2)$$

Now:

- $E(X) = \mu$
- $E(\mu^2) = \mu^2$ since μ is constant

So:

$$\sigma^2 = E(X^2) - 2\mu^2 + \mu^2$$

$$\sigma^2 = E(X^2) - \mu^2$$

✅ Final Result:

$$V(X) = \sigma^2 = E(X^2) - [E(X)]^2$$

where

$$E(X^2) = \sum_x x^2 f(x)$$

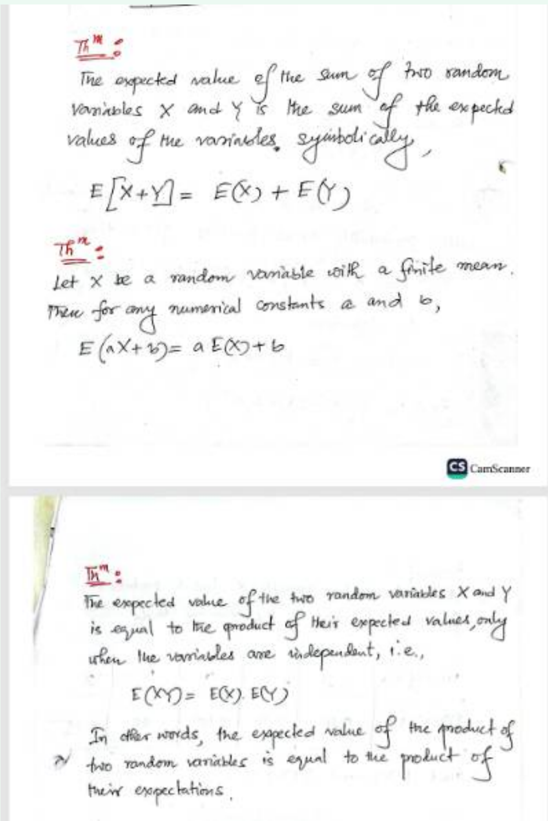
✦ Intuition

- $E(X)$ = the **average** value.
- $V(X)$ = the **average squared distance from the mean**.
- Shortcut formula is useful: instead of calculating deviations from the mean every time, we can just compute $E(X^2)$ and subtract $[E(X)]^2$.

✅ Summary of this page:

- Variance definition: $V(X) = E[(X - \mu)^2]$.
- Standard deviation: $\sigma = \sqrt{V(X)}$.
- Shortcut formula: $V(X) = E(X^2) - [E(X)]^2$.

👉 Do you want me to **work out a full example** (say, variance of a fair die roll) using both the **definition** and the **shortcut formula** so you see the difference in calculation?



Thm :
The expected value of the sum of two random variables X and Y is the sum of the expected values of the variables, symbolically,

$E[X+Y] = E(X) + E(Y)$

Thm :
Let X be a random variable with a finite mean. Then for any numerical constants a and b,

$E(aX+b) = aE(X)+b$

CamScanner

Thm :
The expected value of the two random variables X and Y is equal to the product of their expected values, only when the variables are independent, i.e.,

$E(XY) = E(X) \cdot E(Y)$

In other words, the expected value of the product of two random variables is equal to the product of their expectations.

Great — now you've shared the **properties of expectation**. Let's break each theorem down step by step.

✦ Theorem 1: Linearity of Expectation (Addition Rule)

Statement:
The expected value of the sum of two random variables X and Y is the sum of their expected values:

$$E[X+Y] = E(X) + E(Y)$$

✅ **Key Point:**
This is always true, even if X and Y are dependent.
Expectation is **linear**.

Example:

If $E(X) = 3$ and $E(Y) = 4$, then:

$$E(X+Y) = 3 + 4 = 7$$

✦ Theorem 2: Scaling and Shifting

If X is a random variable with finite mean, and a, b are constants:

$$E(aX + b) = aE(X) + b$$

✅ **Meaning:**

- Multiplying a random variable by a scales its expectation by a.
- Adding a constant b shifts the expectation by b.

Example:

If $E(X) = 5$, then:

$$E(2X + 3) = 2 \cdot 5 + 3 = 13$$

✦ Theorem 3: Expectation of Product (for Independent Random Variables)

If X and Y are **independent** random variables:

$$E(XY) = E(X) \cdot E(Y)$$

✅ **Important Condition:**
This property **only holds if X and Y are independent**. If they are dependent, this equality is not guaranteed.

Example:

- Suppose $E(X) = 2$, $E(Y) = 5$, and X, Y are independent.
  Then:

$$E(XY) = 2 \cdot 5 = 10$$

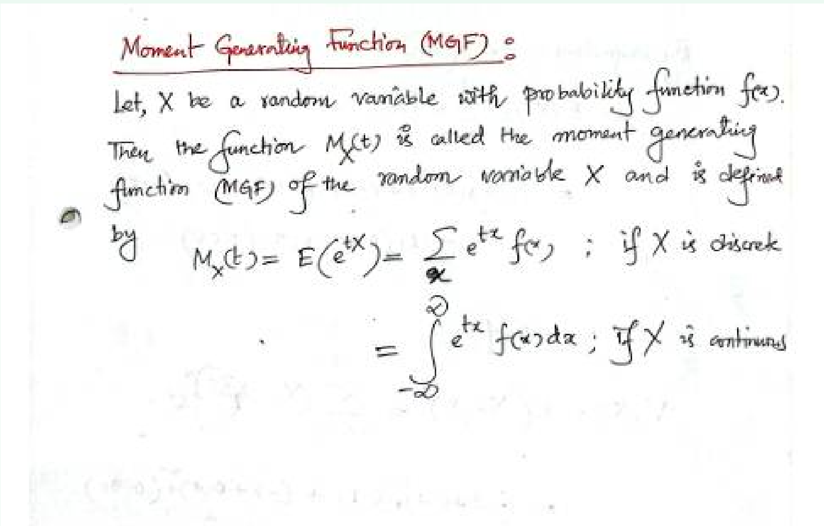
✦ Intuition

- **Linearity:** Expectation is like averaging. When you combine random variables, their averages add up.
- **Scaling:** If you double every outcome, the average doubles. If you shift outcomes, the average shifts.
- **Independence Rule:** For independent random variables, the average of the product is the product of the averages.

✅ **Summary of this page:**

1. $E(X+Y) = E(X) + E(Y)$
2. $E(aX+b) = aE(X) + b$
3. If independent: $E(XY) = E(X) \cdot E(Y)$

👉 Do you want me to also **prove Theorem 1 and Theorem 2 step by step** using the summation definition of expectation, so you see where these rules come from?



Moment Generating Function (MGF):

Let, X be a random variable with probability function f(x).
Then the function M(t) is called the moment generating function (MGF) of the random variable X and is defined by

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x) \; ; \text{ if } X \text{ is discrete}$$
$$= \int_{-\infty}^{\infty} e^{tx} f(x)\,dx \; ; \text{ if } X \text{ is continuous}$$

Give me intuition of what MGF is and what it actually means and what is it used for

Excellent question 👍 — MGFs often look abstract, but they're very powerful once you see the intuition. Let's break it down.

✦ Moment Generating Function (MGF)

Definition (from your note):
For a random variable X with probability function $f(x)$, the **moment generating function** (MGF) is:

- For discrete X:

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x)$$

- For continuous X:

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

✦ Intuition — What does this mean?

1. **Why "moment generating"?**
   - If you take derivatives of $M_X(t)$ at $t = 0$, you get the **moments** of X.
   - For example:
     - First derivative at 0 → $E(X)$ (the mean).
     - Second derivative at 0 → $E(X^2)$.
     - And so on.
       So the MGF "generates" all the moments of the distribution.

---

2. **Connection with averages:**
   - Expectation is like a "weighted average."
   - $M_X(t)$ is the weighted average of the exponential transformation e^{tX}.
   - Because exponentials expand into power series ($e^{tX} = 1 + tX + \frac{(tX)^2}{2!} + \cdots$), the expectation collects all powers of X.

   This is why MGFs are a compact way to encode **all information** about the distribution.

---

3. **Analogy:**
   Think of MGF as a **DNA of a random variable**.

- It captures the entire distribution.
- If two random variables have the same MGF, they have the same distribution (if the MGF exists in a neighborhood around 0).

✦ What is it used for?

1. **Finding moments easily**
   - $E(X) = M'_X(0)$
   - $E(X^2) = M''_X(0)$
   - Variance = $E(X^2) - [E(X)]^2$
2. **Identifying distributions**
   - Many distributions have unique MGFs (Normal, Poisson, Exponential, etc.).
   - If you compute an MGF and it matches a known one, you've identified the distribution.
3. **Sums of independent random variables**
   - If X and Y are independent:

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

   - This property makes MGFs extremely useful in probability proofs, especially in the **Central Limit Theorem**.

✦ Simple Example: Tossing a fair coin

Let X = number of heads in one toss (so $X = 1$ with prob 0.5, and $X = 0$ with prob 0.5).

MGF:

$$M_X(t) = 0.5 \cdot e^{t \cdot 0} + 0.5 \cdot e^{t \cdot 1} = 0.5 + 0.5e^t$$

Now:

- First derivative: $M'_X(t) = 0.5e^t$.
- At $t = 0$: $M'_X(0) = 0.5$. ✅ That's the mean.

- Second derivative: $M_X''(t) = 0.5e^t$.
- At $t = 0$: $M_X''(0) = 0.5$. ✅ That's $E(X^2)$.

So you got moments directly!

✦ Summary

- MGF is the expectation of e^{tX}.
- It encodes **all moments** of a distribution.
- Used to compute mean, variance, higher moments, and to identify distributions.
- Very powerful for analyzing **sums of independent random variables**.

👉 Do you want me to also **show how to derive variance** of a distribution using just the MGF (instead of the standard formulas)? That will demonstrate its real power.

Problem:

A discrete random variable X has a probability function as shown in the following table:

Values of $X: x$	-3	-2	0	1	2
$P(X=x)=f(x)$	0.10	0.30	0.15	0.40	0.05

Find $E(X)$ and $V(X)$.

Sol^n:

By definition we know

$$\mu = E(X) = \sum_{x=-3}^{2} x f(x)$$

$$= (-3)(0.10) + (-2)\cdot(0.30) + (0)(0.15) + (1)(0.40) + (2)(0.05)$$

$$= -0.4$$

$$V(X) = E(X-\mu)^2 = \sum_{x=-3}^{2} (X-\mu)^2 f(x)$$

$$= (-3+0.4)^2 (0.1) + (-2+0.4)^2 (0.30) + (0+0.4)^2 (0.15) + (1+0.4)^2 (0.40) + (2+0.4)^2 (0.05) = 2.54$$

Problem:

Using the formula $V(X) = E(X^2) - \mu^2$ find the variance & standard deviation (SD) of the previous probability function (see previous example)

Sol^n:

By definition we know,

$$E(X^2) = \sum_x x^2 f(x)$$

$$= (-3)^2 \cdot (0.10) + (-2)^2 \cdot (0.30) + (0)^2 \cdot (0.15) + (1)^2 \cdot (0.40) + (2)^2 \cdot (0.05)$$

$$= 2.7$$

Now,

$$V(X) = E(X^2) - \mu^2$$

$$= 2.7 - (-0.4)^2$$

$$= 2.54 \quad \text{Ans.}$$

Hence,

$$SD,\ \sigma = \sqrt{V(X)}$$

$$= \sqrt{2.54}$$

$$= 1.59 \quad \text{(Ans)}$$

Awesome—this example shows both ways to compute the mean and variance. I'll do it cleanly and explain the *why*.

Given PMF

$$x: \quad -3, -2, 0, 1, 2 \qquad p(x): \; 0.10,\; 0.30,\; 0.15,\; 0.40,\; 0.05$$

Quick check: $0.10 + 0.30 + 0.15 + 0.40 + 0.05 = 1$ ✓ (valid PMF)

---

1) Expected value $E(X)$

$$E(X) = \sum_x x\, p(x)$$

$$= (-3)(0.10) + (-2)(0.30) + (0)(0.15) + (1)(0.40) + (2)(0.05)$$

$$= -0.30 - 0.60 + 0 + 0.40 + 0.10 = -0.40$$

So, $\boxed{E(X) = -0.4}$.

Intuition: There's a little more probability mass on the negatives (notably -2 with 0.30), so the long-run average ends up slightly negative (−0.4).

---

2) Variance $V(X)$

Method A — Definition

$$V(X) = E\left[(X-\mu)^2\right], \quad \mu = E(X) = -0.4$$

Compute each term $(x-\mu)^2 p(x)$ and sum:

$$\begin{aligned}
(-3+0.4)^2(0.10) &= (-2.6)^2(0.10) = 6.76(0.10) = 0.676 \\
(-2+0.4)^2(0.30) &= (-1.6)^2(0.30) = 2.56(0.30) = 0.768 \\
(0+0.4)^2(0.15) &= (0.4)^2(0.15) = 0.16(0.15) = 0.024 \\
(1+0.4)^2(0.40) &= (1.4)^2(0.40) = 1.96(0.40) = 0.784 \\
(2+0.4)^2(0.05) &= (2.4)^2(0.05) = 5.76(0.05) = 0.288
\end{aligned}$$

Sum: $0.676 + 0.768 + 0.024 + 0.784 + 0.288 = \boxed{2.54}$.

Method B — Shortcut

First compute $E(X^2) = \sum_x x^2 p(x)$:

$9(0.10) + 4(0.30) + 0(0.15) + 1(0.40) + 4(0.05) = 0.9 + 1.2 + 0 + 0.4 + 0.2 = 2.$

Then

$$V(X) = E(X^2) - [E(X)]^2 = 2.7 - (-0.4)^2 = 2.7 - 0.16 = \boxed{2.54}.$$

Standard deviation

$$\sigma = \sqrt{V(X)} = \sqrt{2.54} \approx \boxed{1.59}.$$

Why these steps make sense

- $E(X)$ is the **weighted average** of outcomes.
- $V(X)$ measures the **average squared distance** from that average.
- The **shortcut** works because expanding $(X - \mu)^2$ and taking expectations yields $E(X^2) - [E(X)]^2$.

Sanity checks

- Mean (−0.4) lies inside the support $[-3, 2]$.
- Variance is positive; SD ~ 1.59 means a "typical" deviation from the mean is about 1.6 units.

If you want, I can put this into a tiny table (x, p(x), x·p(x), x^2·p(x), (x−μ)^2·p(x)) for quick revision.

Problem:

In a coin-tossing game, a man is promised to receive TK. 5 if he gets all heads or all tails when three coins are tossed and he pays off (loses) TK. 3 if either one or two heads appear. How much is he expected to gain in the long run?

Solⁿ:

The random variable X here is the amount of money (in TK.) the man can win. Here, the r.v X will take on a value 5 when the coins show all heads or all tails and -3, otherwise

The table below shows the outcomes of the experiment, value of X and its associated probabilities:

Outcome :	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$X: x$	5	-3	-3	-3	-3	-3	-3	5
$P(X=x)=f(x)$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$

It appears from the above table that the variable X assumes values -3 and $+5$ with probabilities $6/8$ and $2/8$, respectively. Since the value -3 occurs 6 times and 5 occurs 2 times, the expected

Value of X is

$$E(X) = \sum_x x f(x) = -3(6/8) + 5(2/8)$$

$$= -\frac{18}{8} + \frac{10}{8}$$

$$= -1$$

Thus the man is expected to lose TK. 1 in the long run.

Now let us examine what happens if the man receives TK. 5 for all heads or all tails, TK. 0 for 2 heads and pays off TK. 3 for 1 head. Now, the random variable X will take on values 5, 0 and -3 with associated probabilities 2/8, 3/8, and 3/8, respectively.

The expected value in this case will be,

$$E(X) = \sum_x x f(x)$$

$$= 5\left(\frac{2}{8}\right) + 0(3/8) + (-3)\left(\frac{3}{8}\right)$$

$$= \frac{10}{8} + 0 - \frac{9}{8}$$

$$= \frac{1}{8} = 0.125$$

This shows that the man will be marginally gainer winning only 12.5 paisa if the payment is made as designed above.

Nice example—this is a classic "expected gain in a coin-toss game." I'll solve both versions exactly as in your notes and add the intuition + a quick fairness check.

Setup (3 fair coins)

Number of heads $K \sim \mathrm{Binomial}(n = 3, p = 1/2)$.
$\mathbb{P}(K = k) = \binom{3}{k}/8$ for $k = 0, 1, 2, 3$.

$$\mathbb{P}(K=0) = \tfrac{1}{8}, \quad \mathbb{P}(K=1) = \tfrac{3}{8}, \quad \mathbb{P}(K=2) = \tfrac{3}{8}, \quad \mathbb{P}(K=3) = \tfrac{1}{8}.$$

(A) Pay $+5$ for all heads or all tails; pay -3 otherwise

So $X = +5$ when $K = 0$ or 3 (prob $2/8$); $X = -3$ when $K = 1$ or 2 (prob $6/8$).

$$E(X) = 5 \cdot \frac{2}{8} + (-3) \cdot \frac{6}{8} = \frac{10 - 18}{8} = -1.$$

Interpretation: long-run average gain = -1 taka ⇒ the player loses 1 taka per game on average.

(B) Pay $+5$ for all heads or all tails; pay 0 for two heads; pay -3 for one head

Here

$$X = \begin{cases} +5 & (K = 0 \text{ or } 3),\ \mathbb{P} = \frac{2}{8} \\ 0 & (K = 2),\ \mathbb{P} = \frac{3}{8} \\ -3 & (K = 1),\ \mathbb{P} = \frac{3}{8} \end{cases}$$

$$E(X) = 5 \cdot \frac{2}{8} + 0 \cdot \frac{3}{8} + (-3) \cdot \frac{3}{8} = \frac{10 - 9}{8} = \frac{1}{8} = 0.125.$$

Interpretation: long-run average gain = $+0.125$ taka ⇒ the player wins only 12.5 paisa per game on average.

Intuition (why these answers make sense)

- There are 6/8 “middle” outcomes (1 or 2 heads). In version A those are all losses, so the negatives dominate → expected loss.

- In version B, half of the middle mass (2 heads) is neutral (0), which lightens the losses just enough to flip the expectation slightly positive.

(Extra) Fairness checks (optional)

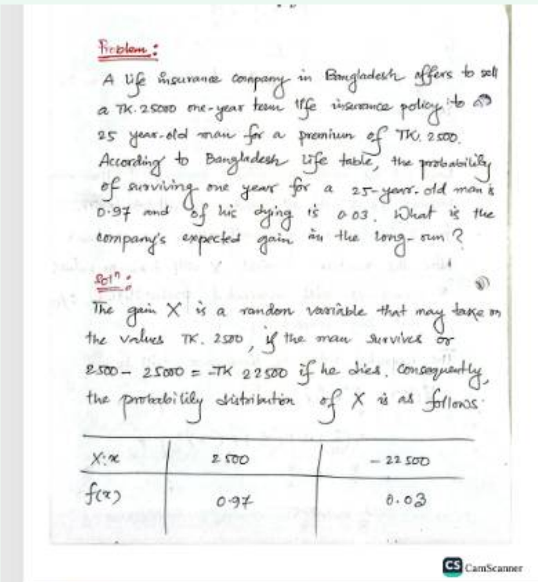
- **Version A**: To make the game fair $(E(X) = 0)$ with payoff $+c$ for all-equal and -3 otherwise:

$$c \cdot \frac{2}{8} - 3 \cdot \frac{6}{8} = 0 \;\Rightarrow\; 2c - 18 = 0 \Rightarrow c = 9.$$

  So paying **Tk. 9** (instead of 5) on all heads/tails makes it fair.

- **Version B**: With payoffs $+5, 0, -3$ it's slightly favorable to the player by 0.125 taka.

If you want, I can tabulate the outcomes and probabilities into a quick revision table for your sheet.



Problem:

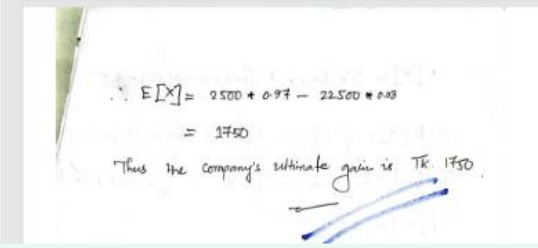
A life insurance company in Bangladesh offers to sell a TK. 25000 one-year term life insurance policy to a 25 year-old man for a premium of TK. 2500. According to Bangladesh life table, the probability of surviving one year for a 25-year-old man is 0.97 and of his dying is 0.03. What is the company's expected gain in the long-run?

Sol^n:

The gain X is a random variable that may take on the values TK. 2500, if the man survives or 2500 − 25000 = −TK 22500 if he dies. Consequently, the probability distribution of X is as follows:

X:x	2500	−22500
f(x)	0.97	0.03

CamScanner



$\therefore E[X] = 2500 * 0.97 - 22500 * 0.03$

$= 1750$

Thus the company's ultimate gain is Tk 1750.

Perfect—this is a **real-life expected value problem** from insurance. Let's carefully unpack it.

Problem restated

- A life insurance company sells a **Tk. 25,000 one-year term policy**.
- Premium charged = **Tk. 2,500**.
- If the person survives → company **keeps the 2,500**.
- If the person dies → company **loses 25,000 – 2,500 = 22,500** (because they collected premium but must pay 25,000).

Probabilities (from life tables):

- Survive = 0.97
- Die = 0.03

Step 1: Define random variable X

Let X = company's gain.

$$X = \begin{cases} +2500 & \text{with probability } 0.97 \\ -22500 & \text{with probability } 0.03 \end{cases}$$

Step 2: Expected value

$$E[X] = 2500(0.97) + (-22500)(0.03)$$

$$= 2425 - 675 = 1750$$

Final Answer

The **expected gain** for the company in the long run is:

$$\boxed{Tk. 1750}$$

Intuition

- Most of the time (97%), the company makes a small profit (+2500).
- Rarely (3%), the company takes a big loss (–22,500).
- Balancing both, on average, the company still gains Tk. 1750 per policy.

This is exactly how insurance companies price policies: they use probability of death/survival × payouts to ensure **positive expected profit**.

👉 Do you want me to also show you how the **break-even premium** (fair premium) can be calculated from this setup (so that $E[X] = 0$)?

Problem:

Let, X denotes the number of spots showing on the face of a well-balanced die after it is rolled once. If $Y = X^2 + 2X$, find

(a) $E[X]$, $E[Y]$, $E[X^2]$, $E[Y^2]$

(b) $V[X]$ and $V[Y]$.

Sol^n:

The random variables X and Y together with their probability distributions are shown in the following table:

$X: x$	1	2	3	4	5	6
$Y: y$	3	8	15	24	35	48
$f(x) = f(y)$	1/6	1/6	1/6	1/6	1/6	1/6

$$E[X] = \sum_x x f(x) = \frac{1}{6}(1+2+3+4+5+6) = 3.5$$

$$\& \ E[Y] = \sum_y y f(y) = \frac{1}{6}(3+8+15+24+35+48) = 22.17$$

To find the variance of X, we find first $E[X^2]$

$$E[X^2] = \sum_x x^2 f(x).$$

$$= \frac{1}{6}(1^2+2^2+3^2+4^2+5^2+6^2)$$

$$\therefore E[X^2] = 15$$

Similarly,

$$E[Y^2] = \sum_y y^2 f(y)$$

$$= \frac{1}{6}(3^2+8^2+15^2+24^2+35^2+48^2)$$

$$\therefore E[Y^2] = 4403$$

Now,

$$\sigma_x^2 = E[X^2] - \{E[X]\}^2$$

$$= 15 - (3.5)^2$$

$$\therefore \sigma_x^2 = 2.75$$

Also,

$$\sigma_y^2 = E[Y^2] - \{E[Y]\}^2$$

$$= 4403 - (22.17)^2$$

$$\therefore \sigma_y^2 = 391.49$$

Perfect—this is a rich problem combining **expected value, variance, and transformations of random variables**. Let's go step by step.

Problem setup

- X = outcome of a fair die → values $1, 2, 3, 4, 5, 6$ each with probability $1/6$.
- Define $Y = X^2 + 2X$.
  So possible Y-values:

$$Y = 3, 8, 15, 24, 35, 48 \quad (\text{when } X = 1, 2, 3, 4, 5, 6)$$

(a) Find $E[X], E[Y], E[X^2], E[Y^2]$

1. $E[X]$

$$E[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

2. $E[Y]$

$$E[Y] = \frac{1}{6}(3 + 8 + 15 + 24 + 35 + 48) = \frac{133}{6} \approx 22.17$$

3. $E[X^2]$

$$E[X^2] = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6} \approx 15.17$$

4. $E[Y^2]$

$$E[Y^2] = \frac{1}{6}(3^2 + 8^2 + 15^2 + 24^2 + 35^2 + 48^2)$$

$$= \frac{1}{6}(9 + 64 + 225 + 576 + 1225 + 2304) = \frac{4403}{6} \approx 733.83$$

(b) Variances

1. Variance of X

$$V(X) = E[X^2] - (E[X])^2 = 15.17 - (3.5)^2 = 15.17 - 12.25 = 2.92$$

2. Variance of Y

$$V(Y) = E[Y^2] - (E[Y])^2 = 733.83 - (22.17)^2 = 733.83 - 491.36 \approx 242.47$$

Final results

- $E[X] = 3.5$
- $E[Y] \approx 22.17$
- $E[X^2] \approx 15.17$
- $E[Y^2] \approx 733.83$
- $V(X) \approx 2.92$
- $V(Y) \approx 242.47$

Intuition

- $E[X] = 3.5$ (the well-known mean of a fair die).
- $Y = X^2 + 2X$ grows quickly (quadratic), so its mean (~22) is much larger.
- Variance of Y is very large (~242) compared to X's variance (~2.9), because squaring “stretches” the distribution, spreading values far from the mean.

👉 Do you want me to also show how to compute $E[Y]$ and $V(Y)$ **directly using formulas of $E[X]$ and $E[X^2]$** (without building the table)? That would be a nice shortcut method.